

Lecture 2 : Calculus and Linear Algebra

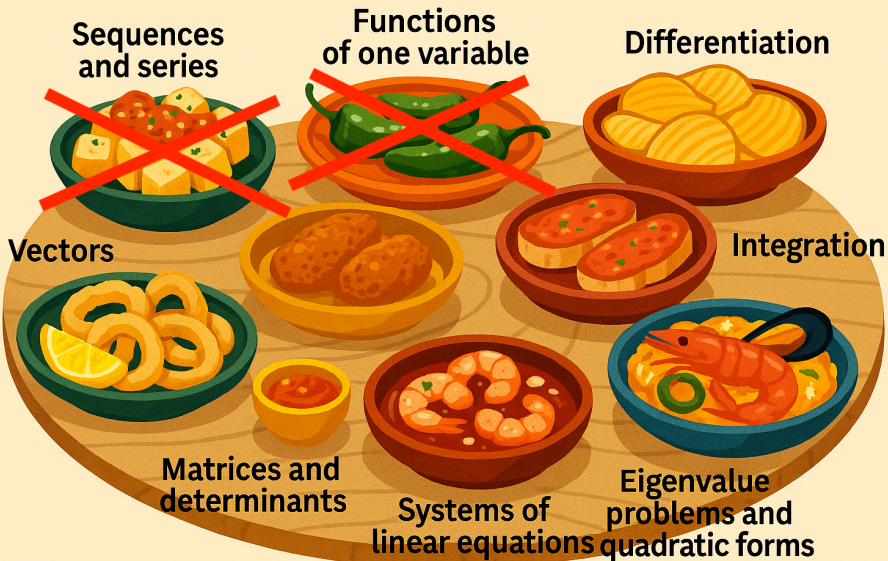
Piotr Zwiernik

Mathematics Brush-up



Barcelona School of Economics

Course content (math tapas)



Optimization in 1D

Monotonicity via derivatives

If f is differentiable on (a, b) :

1. f increasing on $[a, b] \iff f'(x) \geq 0$ on (a, b) .
2. f decreasing on $[a, b] \iff f'(x) \leq 0$ on (a, b) .
3. f constant on $[a, b] \iff f'(x) = 0$ on (a, b) .
4. $f'(x) > 0$ on $(a, b) \Rightarrow f$ strictly increasing on $[a, b]$.
5. $f'(x) < 0$ on $(a, b) \Rightarrow f$ strictly decreasing on $[a, b]$.

Converse of 4-5 can fail at isolated points (e.g., x^3).

Example: $f(x) = \frac{x^3}{3} + 2x^2 + 3x + 1$; $f'(x) = (x+1)(x+3)$ determines monotonicity intervals and local extrema.

Optimal points: first-order conditions

Local optima

Let $f : D \rightarrow \mathbb{R}$. A point $x_0 \in D$ is a **local maximum** if $\exists (a, b) \subset D$ with $x_0 \in (a, b)$ and $f(x) \leq f(x_0)$ for all $x \in (a, b)$. Similarly for **local min** with $\geq$.

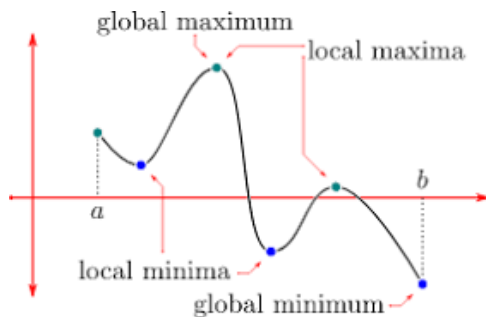
If f has a local optimum at interior x and is differentiable, then $f'(x) = 0$.

To see why this is true, stare at $f(x_0 + h) - f(x_0) \approx f'(x_0)h$.

This condition is **necessary**, not sufficient; see e.g. x^3 .

If x is stationary ($f'(x) = 0$) and f is strictly increasing to the left and strictly decreasing to the right, then x is a local max (reverse for min).

Optimal points



Example: Timing a sale (present value)

Suppose an assets market value is $V(t)$ at time t . At a constant annual interest rate R , the **present value** of selling at time t is

$$P(t) = V(t) e^{-Rt},$$

where e^{-Rt} discounts future cash flows.

Goal: Choose t to maximize $P(t)$.

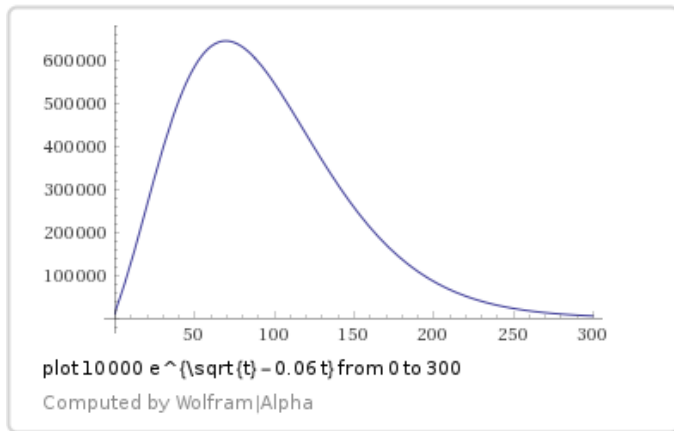
FOC (set derivative to zero):

$$P'(t) = V'(t)e^{-Rt} - R V(t)e^{-Rt} = 0 \quad \Rightarrow \quad \frac{V'(t)}{V(t)} = R.$$

Interpretation: Sell when the assets *instantaneous percentage growth rate* (matches) the *interest rate you could earn in the bank*.

Example: If $V(t) = 10000 e^{\sqrt{t}}$ and $R = 6\%$, then $P(t) = 10000 e^{\sqrt{t} - 0.06 t}$, which is maximized near $t \approx 69.44$.

Plot of the present value $P(t)$



To certify a **global** optimum on an interval, compare values at boundaries and critical points. Here $P(0) = 10000$, $\lim_{t \rightarrow \infty} P(t) = 0$.

Second-order conditions

Test: If $f'(x_0) = 0$ and $f''(x_0) < 0$ (> 0), then x_0 is a strict local max (min).

Indeed,
$$f(x_0 + h) \approx f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 = f(x_0) + \frac{1}{2}f''(x_0)h^2.$$

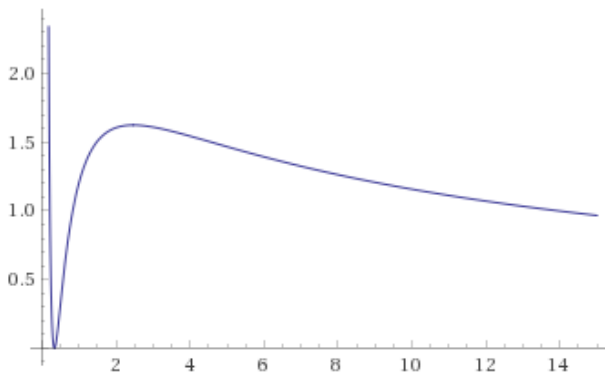
Example: For $f(x) = \frac{\log^2(3x)}{x}$ ($x > 0$):

$$f'(x) = \frac{(2 - \log 3x) \log 3x}{x^2}, \quad f''(x) = \frac{2(1 - 3 \log 3x + \log^2 3x)}{x^3}.$$

Critical points at $x_1 = \frac{1}{3}$ (min), $x_2 = \frac{e^2}{3}$ (max).

Question: What if $f''(x_0) = 0$? (assume all derivatives exist)

Plot of $f(x) = \frac{\log^2 3x}{x}$, $x > 0$



plot $\ln^2(3x)/x$ from 0 to 15 | Computed by Wolfram|Alpha

$\lim_{x \rightarrow \infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $f(x_1) = 0$, $f(x_2) > 0 \Rightarrow x_1$ global min,
no global max.

Convexity and concavity

If f is twice differentiable on (a, b) :

1. f convex on $[a, b] \Leftrightarrow f''(x) \geq 0$ for all $x \in (a, b)$.
2. f concave on $[a, b] \Leftrightarrow f''(x) \leq 0$ for all $x \in (a, b)$.
3. $f'' > 0$ (< 0) $\Rightarrow f$ strictly convex (concave).

Theorem: If f and g are convex (or concave), then $f \circ g$ is convex (or concave).

Example: $f(x) = e^{x^2}$ is strictly convex on $[0, \infty)$.

Economics:

- **Cost functions:** usually convex — producing extra units becomes progressively more expensive (e.g. overtime wages, machine wear).
- **Utility functions:** often concave — reflects diminishing marginal utility (the first slice of pizza gives more satisfaction than the tenth).
- **Machine learning:** convex losses (e.g. squared error, logistic loss + ℓ_2 regularization) ensure optimization problems are tractable and stable.

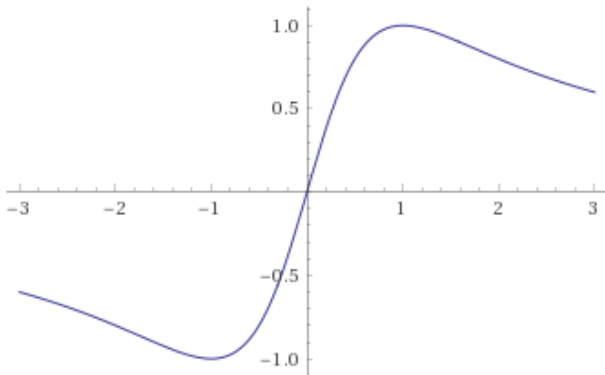
Example of changing curvature

Let $f(x) = \frac{2x}{x^2 + 1}$. Then

$$f'(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}, \quad f''(x) = \frac{4(x^3 - 3x)}{(x^2 + 1)^3}.$$

So f is strictly convex on $[-\sqrt{3}, 0] \cup [\sqrt{3}, \infty)$ and strictly concave on $(-\infty, -\sqrt{3}] \cup [0, \sqrt{3}]$.

Plot of $f(x) = \frac{2x}{x^2 + 1}$



plot $2x/(x^2+1)$ from -3 to 3 | Computed by Wolfram|Alpha

Inflexion points at $x \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Global max at $x = 1$, min at $x = -1$, and $\lim_{|x| \rightarrow \infty} f(x) = 0$.

Inflexion points

If f is C^n on (a, b) , an **inflexion point** at x_0 occurs when curvature changes sign. A sufficient test:

$$f''(x_0) = \dots = f^{(n-1)}(x_0) = 0, \quad f^{(n)}(x_0) \neq 0, \quad n \text{ odd.}$$

Example: $f(x) = x^3$ has $f''(0) = 0$, $f^{(3)}(0) = 6 \neq 0 \Rightarrow$ inflexion at 0.

Taylor series expansion

Mean-value theorem

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (\text{equivalently } f(b) = f(a) + f'(c)(b - a)).$$

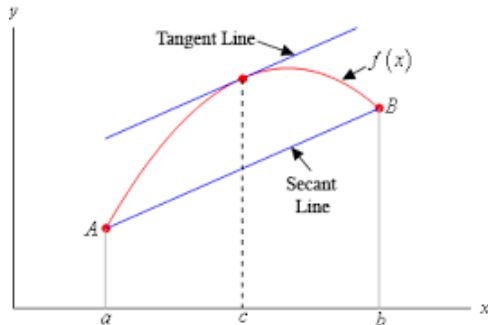
Proof idea: consider $h(x) = (f(b) - f(a))x - (b - a)f(x)$; since $h(a) = h(b)$, h has an interior extremum.

Interpretation: the instantaneous slope matches the secant slope at some point.

Geometry of the mean-value theorem

Recall: $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Taylor formula

Theorem (Taylor's formula of order n): Let f be $n + 1$ times differentiable on (a, b) , and let $x_0 \in (a, b)$ given. Then, for any $x \in (a, b)$, if $n = 1$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{error}.$$

If $n \geq 2$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \text{error}.$$

The error is small if x is close to x_0 , therefore, this theorem says that differentiable functions may be **locally approximated by a polynomial** called the **Taylor polynomial**.

Economics/data: local linear/quadratic approximations underlie elasticity (log-linearization), Delta method, and second-order price responses.

Taylor remainder and a handy expansion

Remainder (Lagrange form)

$$R_n^y(x) = \frac{f^{(n+1)}(y)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } y \text{ between } x_0 \text{ and } x.$$

If $|f^{(n+1)}| \leq M$ near x_0 , then $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$.

Example (log around 1): With $x_0 = 1$,

$$\log x \approx (x - 1) - \frac{1}{2}(x - 1)^2, \quad R_2^y(x) = \frac{1}{3y^3}(x - 1)^3.$$

Finance quick check: For small r , $\log(1 + r) \approx r - \frac{r^2}{2} \Rightarrow \text{EAR} \approx \text{APR} - \frac{1}{2}\text{APR}^2$ (small correction).

Short comment about small-o notation

Writing $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$ we conclude

$$\frac{R_n^y(x)}{|x - x_0|^n} \leq \frac{M}{(n+1)!} |x - x_0| \text{ and so}$$

$$\lim_{x \rightarrow x_0} \frac{R_n^y(x)}{|x - x_0|^n} = 0.$$

This we write as $R_n^y(x) = o(|x - x_0|^n)$.

In particular: If f is twice differentiable:

$$f(x+h) = f(x) + f'(x)h + o(|h|).$$

The generalization of this last expression to multivariate functions will provide powerful insights into their optimization.

Chapter 4: Integration

Often a function f is given and we are looking for F whose derivative is f .

Example: the marginal cost function $C'(x)$ is known (how cost changes with production x). We want the cost $C(x)$ itself.

Idea: **integration** reverses differentiation: it accumulates small changes.

Read Chapter 5 of Werner-Sotskov **Exercises** 5.1(a)-(b), 5.2(a)-(b), 5.3(c)

Indefinite integrals

Indefinite integrals

A differentiable F is an **antiderivative** of f if $F'(x) = f(x)$ on a common domain.

Fact: all antiderivatives differ by a constant: if $F'(x) = f(x)$, then any $\tilde{F}(x) = F(x) + c$ also satisfies $\tilde{F}'(x) = f(x)$.

Definition: the **indefinite integral** is

$$\int f(x) dx = F(x) + c.$$

Linearity:

1. $\int (f + g) dx = \int f dx + \int g dx$
2. $\int c f dx = c \int f dx$

Indefinite integrals you should know

Templates:

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$$

$$2. \int \frac{1}{x} dx = \log |x| + c$$

$$3. \int e^x dx = e^x + c$$

$$4. \int \sin x dx = -\cos x + c$$

$$5. \int \cos x dx = \sin x + c$$

$$6. \int a^x dx = \frac{a^x}{\log a} + c \quad (a > 0, a \neq 1)$$

Always check by differentiating the right-hand side.

Integration by substitution

Theorem (Substitution)

If $t = g(x)$ and F is an antiderivative of f , then

$$\int f(g(x)) g'(x) dx = \int f(t) dt = F(t) + c = F(g(x)) + c.$$

Examples:

$$1. \int (ax + b)^n dx = \frac{1}{a} \int t^n dt = \frac{(ax + b)^{n+1}}{a(n+1)} + c.$$

($t = ax + b$, so $dt = a dx$.)

$$2. \int \frac{e^x}{\sqrt[3]{1 + e^x}} dx = \int t^{-1/3} dt = \frac{3}{2} t^{2/3} + c = \frac{3}{2} (1 + e^x)^{2/3} + c.$$

($t = 1 + e^x$, so $dt = e^x dx$.)

Integration by parts

Theorem (Integration by parts)

For differentiable u, v ,

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

Proof: Differentiate $u(x)v(x)$ and integrate.

Example: with $u = \log x$, $v' = 1$,

$$\int \log x dx = x \log x - \int 1 dx = x(\log x - 1) + c.$$

A combo: substitution then parts

Compute $\int \sin \sqrt{x} \, dx$.

Substitute $t = \sqrt{x}$, so $x = t^2$ and $dx = 2t \, dt$:

$$\int \sin \sqrt{x} \, dx = 2 \int t \sin t \, dt.$$

Now parts with $u = t$, $v' = \sin t$:

$$2 \int t \sin t \, dt = 2 \left(-t \cos t + \int \cos t \, dt \right) = 2(-t \cos t + \sin t) + c.$$

Back-substitute $t = \sqrt{x}$:

$$\boxed{\int \sin \sqrt{x} \, dx = 2 \left(-\sqrt{x} \cos \sqrt{x} + \sin \sqrt{x} \right) + c}.$$

A combo: substitution then parts



Definite integrals

Definite integral as area and accumulation

For continuous $f : [a, b] \rightarrow \mathbb{R}$ with $f \geq 0$, the definite integral

$$\int_a^b f(x) dx$$

is the area under f between a and b . More generally it accumulates signed change.

Properties:

1. $\int_b^a f = -\int_a^b f$
2. $\int_a^b cf = c \int_a^b f$
3. If $c \in [a, b]$, then $\int_a^b f = \int_a^c f + \int_c^b f$
4. $\left| \int_a^b f \right| \leq \int_a^b |f|$
5. If $f \leq g$ on $[a, b]$, then $\int_a^b f \leq \int_a^b g$

Some uses: cumulative revenue from a known marginal revenue curve, energy used by a device with power draw $P(t)$, or probability mass from a density.

Fundamental Theorem of Calculus

Fundamental Theorem of Calculus

If f is continuous on $[a, b]$ and F is an antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Moreover, $G(t) = \int_a^t f(x) dx$ is differentiable and $G'(t) = f(t)$.

Marginal to total: If $C'(x)$ is marginal cost, then the change in total cost from $x = 300$ to $x = 400$ is

$$C(400) - C(300) = \int_{300}^{400} C'(x) dx.$$

Example: $C'(x) = 6 - \frac{60}{x+1}$ for $x \in [0, 1000]$:

$$\int_{300}^{400} \left(6 - \frac{60}{x+1}\right) dx = \left(6x - 60 \log |x+1|\right)_{300}^{400} \approx 582.79.$$

Application: proving $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ via integrals

Define $\log x = \int_1^x \frac{1}{t} dt$. Let e^x be the inverse of $\log x$; then $1 = \int_1^e \frac{1}{t} dt$.

For $t \in [1, 1 + \frac{1}{n}]$,

$$\frac{1}{n+1} = \int_1^{1+\frac{1}{n}} \frac{1}{1+\frac{1}{n}} dt \leq \int_1^{1+\frac{1}{n}} \frac{1}{t} dt \leq \int_1^{1+\frac{1}{n}} 1 dt = \frac{1}{n}.$$

So

$$\frac{1}{n+1} \leq \log\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}.$$

Exponentiate and rearrange to obtain

$$\frac{e}{1 + \frac{1}{n}} \leq \left(1 + \frac{1}{n}\right)^n \leq e,$$

and let $n \rightarrow \infty$.



Application: Taylor with integral remainder (up to $n = 2$)

Apply FTC repeatedly:

$$\begin{aligned}f(x) &= f(x_0) + \int_{x_0}^x f'(t_1) dt_1 \\&= f(x_0) + f'(x_0)(x - x_0) + \int_{x_0}^x \int_{x_0}^{t_1} f''(t_2) dt_2 dt_1 \\&= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \int_{x_0}^x \int_{x_0}^{t_1} \int_{x_0}^{t_2} f^{(3)}(t_3) dt_3 dt_2 dt_1.\end{aligned}$$

By the intermediate value theorem there is y between x_0 and x with

$$\int \int \int f^{(3)}(t_3) dt_3 dt_2 dt_1 = \frac{f^{(3)}(y)}{3!}(x - x_0)^3.$$

Chapter 5: Vectors

A **vector** is an ordered n -tuple of real numbers: $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

Why care in econ/data:

- a bundle of n goods (quantities),
- a user's features or a product's attributes,
- a portfolio's weights across n assets,
- a document's word counts or an embedding.

Read Chapter 6 of Werner-Sotskov; Simon-Blume Chs. 10-11.

Exercises 6.2, 6.3, 6.4, 6.6, 6.7, 6.8

Vectors and the Euclidean space

Definition and notation

A **vector** $\mathbf{v}$ is an ordered n -tuple $(v_1, \dots, v_n)$ of real numbers called **coordinates**.

Notation: $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

The zero vector is $\mathbf{0} = (0, \dots, 0)$.

The i -th unit vector is $\mathbf{e}_i$ (a 1 in position i , zeros elsewhere).

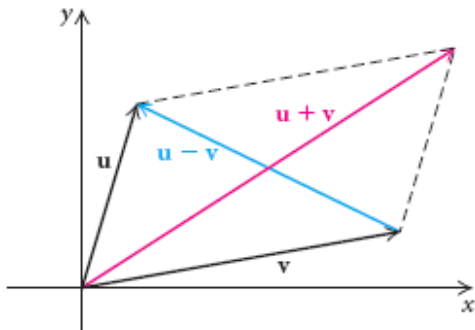
Operations:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n), \quad \lambda \mathbf{v} = (\lambda v_1, \dots, \lambda v_n).$$

These satisfy the usual commutative, associative, and distributive laws.

Sum and difference of two vectors

Note $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v}$.



Inner product and norm

The **inner product** (dot product) of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i.$$

The **Euclidean norm** is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Properties:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
3. **Cauchy-Schwarz:** $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$
4. **Triangle inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Example (platform revenue): hours watched per genre $\mathbf{u} = (500, 200, 50)$ and euro-per-hour rates $\mathbf{v} = (2, 3, 5)$ yield total revenue $\langle \mathbf{u}, \mathbf{v} \rangle = 2150$.

The Law of Cosines and angles

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\| \|\mathbf{v}\| \cos(\angle(\mathbf{u}, \mathbf{v})).$$

Equivalently,

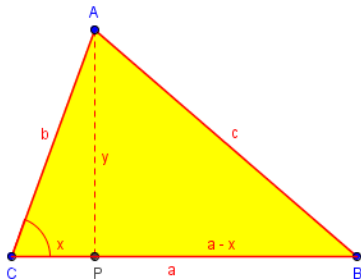
$$\cos(\angle(\mathbf{u}, \mathbf{v})) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

We prove that $c^2 = a^2 + b^2 - 2ab \cos(C)$:

By Pythagoras, $x^2 + y^2 = b^2$ and $(a-x)^2 + y^2 = c^2$.

Subtract to eliminate y^2 to get $c^2 = a^2 + b^2 - 2ax$.

Use $\cos C = x/b$.



Orthogonality and linear independence

Orthogonality

Definition: $\mathbf{u} \perp \mathbf{v}$ if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Angle example: For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (5, -4, -1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 11, \quad \cos \angle(\mathbf{u}, \mathbf{v}) = \frac{11}{\sqrt{14}\sqrt{42}} \approx 0.4537,$$

so the angle is about 63° .

Geometric test: $\mathbf{u} \perp \mathbf{v}$ iff $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Cauchy-Schwarz via completing the square

Assume nonzero $\mathbf{u}, \mathbf{v}$. For any real t ,

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2.$$

Complete the square in t :

$$\|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2 = \|\mathbf{v}\|^2 \left(t - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \right)^2 + \left(\|\mathbf{u}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{v}\|^2} \right).$$

Since the left side is ≥ 0 for all t , the second term must be ≥ 0 :

$$\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2.$$

Equality condition: equality holds iff $\|\mathbf{u} - t\mathbf{v}\|^2 = 0$ for $t = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, i.e., $\mathbf{u} = t\mathbf{v}$ (collinear vectors).



Linear dependence, independence, and bases

Linear combination: $\mathbf{u} = \sum_{i=1}^m \lambda_i \mathbf{v}_i$.

Linear independence: $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent if

$$\sum_{i=1}^m \lambda_i \mathbf{v}_i = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_m = 0.$$

Basis: Any n linearly independent vectors in $\mathbb{R}^n$ form a basis. Then every $\mathbf{u} \in \mathbb{R}^n$ has a **unique** representation $\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$.

Standard basis: $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

Subspaces, span, and dimension

For $E = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, the **span** is

$$\text{span}(E) = \left\{ \sum_{i=1}^k \lambda_i \mathbf{v}_i : \lambda_i \in \mathbb{R} \right\}.$$

A **subspace** $V \subset \mathbb{R}^n$ is any span. A **basis of V** is a linearly independent set that spans V . The **dimension** of V is the size of any basis.

Exercise: Basis of $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

Orthogonal complement and projection

Orthogonal complement

For a subspace $V \subset \mathbb{R}^n$,

$$V^\perp = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle = 0 \ \forall \mathbf{v} \in V\}.$$

Projection theorem

For any $\mathbf{y} \in \mathbb{R}^n$ and subspace V , there is a unique $\hat{\mathbf{y}} \in V$ with $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. We call $\hat{\mathbf{y}}$ the orthogonal projection of $\mathbf{y}$ onto V .

Least squares view: With design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and response $\mathbf{y}$, the LS fit $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$.

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$



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Chapter 6: Matrices and determinants

A **matrix** $A \in \mathbb{R}^{m \times n}$ is a table with m rows, n columns. The (i, j) entry is a_{ij} .

We recall basic facts about matrices.

Why care: matrices express linear maps, data tables, network flows, input-output models, regressions, transformations, and more.

Read Werner-Sotskov Ch. 7; Simon-Blume Chs. 8-9.

Exercises 7.6, 7.9(b,c,d), 7.12, 7.14(a), 7.16, 7.18

Matrices

A table of numbers with m rows and n columns: $A \in \mathbb{R}^{m \times n}$. The (i, j) -th entry is denoted by a_{ij} .

Special matrices:

- zero matrix: $\mathbf{0}_{m \times n} \in \mathbb{R}^{m \times n}$
- identity matrix: $\mathbb{I}_n \in \mathbb{R}^{n \times n}$.
- **transposition**: $A^T \in \mathbb{R}^{n \times m}$, $(A^T)_{ij} = a_{ji}$.
- square matrix: $m = n$.
- symmetric matrix: square matrix such that $A = A^T$.
- diagonal matrix: $a_{ij} = 0$ if $i \neq j$.
- lower (upper) triangular: $a_{ij} = 0$ if $i < j$ ($i > j$).
- $\mathbf{v} \in \mathbb{R}^n$ is treated as $n \times 1$ matrix, $\mathbb{R}^n \simeq \mathbb{R}^{n \times 1}$.

Basic matrix operations

Addition and scalar multiplication are entrywise:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

They satisfy the usual commutative, associative, and distributive laws.

Matrix product: If $A \in \mathbb{R}^{m \times p}$ and $B \in \mathbb{R}^{p \times n}$, then $C = AB \in \mathbb{R}^{m \times n}$ with

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}.$$

Example:

$$\begin{pmatrix} 2 & 3 & 4 & 1 \\ 7 & -1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 7 \\ 3 & -1 \\ 4 & 0 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 30 & 15 \\ 15 & 66 \end{pmatrix}.$$

Algebraic properties

Facts:

1. $(AB)C = A(BC)$
2. $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
3. Generally **not** commutative: $AB \neq BA$
4. $AI_n = I_m A = A$ (dimensions must match)
5. $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(\lambda A)^T = \lambda A^T$, $(AB)^T = B^T A^T$

Remark: AA^T is always symmetric.

Matrix times vector

We treat $\mathbb{R}^n$ as column vectors $\mathbb{R}^{n \times 1}$. If $A \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^n$ then $A\mathbf{x} \in \mathbb{R}^m$ is a linear combination of the columns of A with coefficients x_i :

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

e.g. **Another look at the LS method:** $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$

$$\text{minimize}_{\beta \in \mathbb{R}^d} \quad \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Finds the closest point to $\mathbf{y}$ in the space spanned by the columns of $\mathbf{X}$.

Orthogonal projection

Theorem: Given a vector $\mathbf{y} \in \mathbb{R}^n$ and a subspace $V \subset \mathbb{R}^n$ there exists a unique $\hat{\mathbf{y}} \in V$ such that $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. Let $\mathbf{x}_1, \dots, \mathbf{x}_d$ be a basis of $V \subset \mathbb{R}^n$.

$\mathbf{X} \in \mathbb{R}^{n \times d}$ with columns $\mathbf{x}_1, \dots, \mathbf{x}_d$.

$\hat{\mathbf{y}} \in V$ means $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\lambda}$ for some $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$.

$\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$ means $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$.

The unique solution: $\boldsymbol{\lambda} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ giving

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Matrix inverse

A square matrix A is **invertible** if there exists a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

We want to show:

$$A \in \mathbb{R}^{n \times n} \text{ is invertible} \iff A\mathbf{x} = \mathbf{0}_n \text{ only for } \mathbf{x} = \mathbf{0}_n.$$

Two observations:

1. $A\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$ iff columns of A are lin. independent.
2. n independent vectors in $\mathbb{R}^n$ form a basis and so $\forall i = 1, \dots, n \exists \mathbf{b}_i$ such that $A\mathbf{b}_i = \mathbf{e}_i$.
3. This gives $B \in \mathbb{R}^{n \times n}$ such that $AB = \mathbb{I}_n$.

This is enough to show that the matrix $\mathbf{X}^\top \mathbf{X}$ on slide 46 is invertible.

Note: If $(\mathbf{X}^\top \mathbf{X})\mathbf{x} = \mathbf{0}$ then $\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})\mathbf{x} = 0$ but this only possible if $\mathbf{x} = \mathbf{0}$.

Important spaces and rank

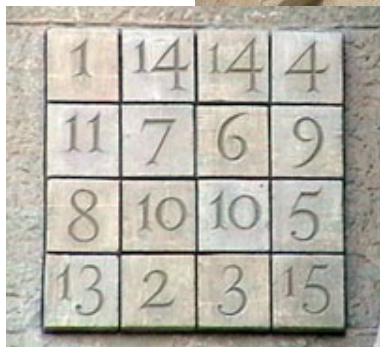
For $A \in \mathbb{R}^{m \times n}$:

- **Column space** (image) $\text{Im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subset \mathbb{R}^m$
- **Row space** $\text{Im}(A^T) \subset \mathbb{R}^n$
- **Kernel** (null space) $\ker(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} \subset \mathbb{R}^n$
- **Rank** $\text{rank}(A) = \dim \text{Im}(A) = \dim \text{Im}(A^T)$

Orthogonality: Row space is orthogonal to $\ker(A)$.

Rank-nullity: $\text{rank}(A) + \dim \ker(A) = n$.

Try: $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.



A close-up photograph of a 4x4 grid of numbers carved into a stone surface. The numbers are arranged in four rows and four columns. The first row contains 1, 14, 14, and 4. The second row contains 11, 7, 6, and 9. The third row contains 8, 10, 10, and 5. The fourth row contains 13, 2, 3, and 15.

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15



Orthogonal matrices

An $n \times n$ matrix A is called **orthogonal** if $AA^\top = I_n$.

Remark: If A is orthogonal, its row vectors (and also its column vectors) are pairwise orthogonal unit vectors.

Proof: Let $\mathbf{r}_i$ and $\mathbf{r}_j$ be the i -th and j -th rows of A . The (i, j) entry of $AA^\top$ is the scalar product $\mathbf{r}_i^\top \mathbf{r}_j$. If $AA^\top = I$, then $\mathbf{r}_i^\top \mathbf{r}_j = 0$ for $i \neq j$ (orthogonality) and $\mathbf{r}_i^\top \mathbf{r}_i = \|\mathbf{r}_i\|^2 = 1$ (unit length). The same holds for columns using $A^\top A = I_n$.

Example:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Singular value decomposition (SVD)

Singular value decomposition

For every $A \in \mathbb{R}^{m \times n}$ there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ and a “diagonal” matrix $\Lambda \in \mathbb{R}^{m \times n}$ such that

$$A = U\Lambda V^T.$$

If $m \leq n$ then Λ has m “diagonal” entries.

We can assume $\Lambda_{11} \geq \dots \geq \Lambda_{mm} \geq 0$.

These diagonal entries are called **singular values** of A .

Spectral decomposition

Singular value decomposition

Suppose $A \in \mathbb{R}^{n \times n}$ is symmetric. Then there exist orthogonal $U \in \mathbb{R}^{n \times n}$ and a diagonal Λ such that

$$A = U\Lambda U^{\top}.$$

Denote $(\lambda_1, \dots, \lambda_n) = \text{diag}(\Lambda)$.

We take $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$.

Elementary matrix operations

Column or row operations:

1. **Swap** two columns (or rows)
2. **Scale** a column (or row) by $\lambda \neq 0$
3. **Add** a multiple of one column (or row) to another

Each is implemented by multiplying by a suitable elementary matrix on the right (for column ops) or left (for row ops). Useful for Gaussian elimination and determinant computation.

To see what is happening, let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and multiply from the right by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}$, or $\begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}$.

Determinants: definition

For $A \in \mathbb{R}^{n \times n}$, let A_{ij} be the submatrix with row i and column j removed. Define recursively

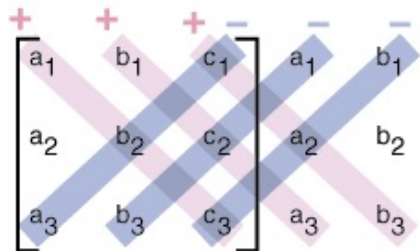
$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}), \quad \det([a_{11}]) = a_{11}.$$

For $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$

For $n = 3$:

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

A memory aid for $n = 3$



$$\det A = (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) - (a_3 b_2 c_1 + b_3 c_2 a_1 + c_3 a_2 b_1)$$

Cofactor expansion and properties

Cofactor expansion: expand by any row i or any column j :

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Properties:

1. $\det(A) = \det(A^T)$
2. If A is triangular, $\det(A)$ is the product of diagonal entries
3. $\det(AB) = \det(A) \det(B)$
4. Swapping two rows (or columns) flips the sign of $\det$
5. Scaling a row (or column) by λ scales $\det$ by λ
6. Adding a multiple of one row to another leaves $\det$ unchanged
7. $\det(A) = 0$ iff rows (or columns) are linearly dependent

Determinants by elimination

Use row operations (keeping track of determinant changes) to reach triangular form.

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & 3 \end{vmatrix} = (-6) \cdot 3 = -18.$$

First: $r_2 \leftarrow r_2 - 3r_1$. Then: $r_3 \leftarrow r_3 - r_1$.

Geometric meaning: $|\det(A)|$ is the area/volume scaling of the linear map $x \mapsto Ax$ (and its sign encodes orientation).

Linear systems and Cramer's rule

A system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{n \times n}$, unknown $\mathbf{x} \in \mathbb{R}^n$, and data $\mathbf{b}$. If A is **nonsingular** ($\det A \neq 0$), the solution is unique.

Cramer's rule: Let $A_j(\mathbf{b})$ be A with column j replaced by $\mathbf{b}$. Then

$$x_j = \frac{\det A_j(\mathbf{b})}{\det A}, \quad j = 1, \dots, n.$$

Note: Great for theory, not used for large-scale computation.

Example:

$$A = \begin{pmatrix} 2 & 3 & 5 \\ 1 & 0 & 2 \\ -1 & -4 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = \left(\frac{75}{8}, -\frac{63}{16}, -\frac{19}{16} \right).$$

Linear mappings

A mapping $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A(\mathbf{x}_1) + A(\mathbf{x}_2), \quad A(\lambda \mathbf{x}) = \lambda A(\mathbf{x}).$$

Then there exists an $m \times n$ matrix (also denoted A) with $A(\mathbf{x}) = A\mathbf{x}$.

Columns as images: $A(\mathbf{e}_i) = \mathbf{a}_i$ (the i -th column), and $A(\mathbf{x}) = \sum_i x_i A(\mathbf{e}_i)$.

Examples: scalings, rotations, reflections, projections, feature maps in ML, Leontief input-output in economics.

Two simple linear maps

1. Reflection across the y -axis:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}.$$

2. Rotation by 45° counterclockwise:

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Check the images of $\mathbf{e}_1$ and $\mathbf{e}_2$ to see the action.

Inverse matrix and properties

A square A is invertible if A^{-1} exists with $AA^{-1} = A^{-1}A = I_n$.

Properties:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(\lambda A)^{-1} = \lambda^{-1}A^{-1}$ for $\lambda \neq 0$
5. $\det(A^{-1}) = 1/\det(A)$

Solve $A\mathbf{x} = \mathbf{b}$ by $\mathbf{x} = A^{-1}\mathbf{b}$ when A is invertible.

Computing inverses

Cofactor formula: If A is nonsingular,

$$A^{-1} = \frac{1}{\det A} C(A)^T, \quad C(A)_{ij} = (-1)^{i+j} \det(A_{ij}).$$

For 2×2 :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

For larger n , numerical methods use elimination (LU), not cofactors.

Example: Input-output model

Let a_{ij} be units of good i needed to produce 1 unit of good j . Put $A = (a_{ij})$. Let $\mathbf{x}$ be total output and $\mathbf{y}$ the vector of final demand.

Accounting identity: output = internal demand + final demand

$$\mathbf{x} = A\mathbf{x} + \mathbf{y} \quad \Leftrightarrow \quad (I_n - A)\mathbf{x} = \mathbf{y} \quad \Rightarrow \quad \mathbf{x} = (I_n - A)^{-1}\mathbf{y}$$

provided $I_n - A$ is invertible.

Interpretation: $(I - A)^{-1} = I + A + A^2 + \dots$ accumulates direct, indirect, and higher-order input needs when it converges.

A triangular example

Theorem: If A is strictly upper triangular (zeros on and below diagonal), then $A^n = 0$ and

$$(I_n - A)^{-1} = I_n + A + A^2 + \cdots + A^{n-1}.$$

Check $(I - A)(I + A + \cdots + A^{n-1}) = I - A^n = I$.

Example:

$$A = \begin{pmatrix} 0 & 3 & 5 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = (66, 15, 4).$$